

EFY Note

The source code of this project is available for free download at www.electronicsforu.com

DO-IT-YOURSELF: PROJECT

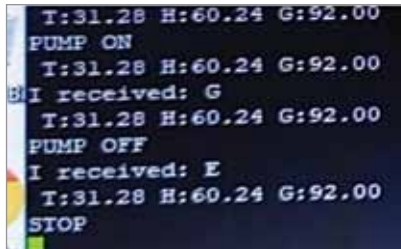


Fig. 6: Software command seen on LCD

Step 2. Complete wiring of the project as per circuit diagram shown in Fig. 3

Step 3. For sensor's reading, the sensor's output is connected to an Arduino analogue input. So, the Arduino will read the input data from the sensors like humidity and temperature of the room.

Step 4. Convert the data read from the sensor into normalised data set to show the data as a percentage.

Step 5. Log into an SD card if you want to store data. (It is not compulsory.) By using the SD

library, we can store our data like temperature and humidity on an SD card. The maximum capacity is 2GB for SD cards and 16GB for micro SD cards.

Step 6. We can also send data from the serial monitor to Arduino by entering text in our system. When we command Forward, it will move in the forward direction. Likewise, Backward, Right, and Left commands move it in backward, right, and left directions, respectively.

After properly assembling the robot, as shown in Fig. 1, we may power the robot. The robot can now be controlled by sending the commands using the serial terminal for forward, backward, and other movements set in code. The robot can now enter hazardous environments to provides various gas data.

This robot provides the ambient temperature and detects any flame in the coal mine and extinguishes it. Using Zigbee wireless technology, the LCD connected to the robot (see Fig. 5) shows the values of temperature, humidity, and gas presence in the coal mine. Fig. 6 shows the software commands on the LCD. **EFY**



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